SQL Interview Questions

Navigation Product Analytics — 25 SQL Challenges

Based on tables: users, navigation_sessions, route_options, user_events, experiments, retention_daily

Format: Difficulty-tagged, no solutions provided

Goal: Answer the Key Product Questions from the business case using only SQL

How to use this document

Each question references the schema in data_dictionary.xlsx. Solutions are intentionally not included — write and validate each query against the provided CSVs (e.g., load them into SQLite, DuckDB, PostgreSQL, or BigQuery). Questions are ordered roughly from foundational to advanced within each topic area, not strictly by difficulty.

Q1 — Easy | JOIN / Aggregation

For each city, calculate the total number of navigation sessions and the average distance_km. Order by session count descending.

Q2 — Easy | CASE WHEN

Write a query that labels each session as 'Accepted' or 'Rejected' by comparing selected_route_id to recommended_route_id, then count sessions in each label.

Q3 — Easy | GROUP BY

Calculate the recommended-route acceptance rate by recommended_route_type.

Q4 — Easy | Filtering / Aggregation

Find the average actual_duration_minutes and average recommended_eta_minutes for sessions where traffic_level = 'High', excluding rows with NULL actual_duration_minutes.

Q5 — Easy | JOIN

Join navigation_sessions to users and report the acceptance rate split by user_type (New vs Returning).

Q6 — Easy | GROUP BY / HAVING

List cities where the average traffic_uncertainty_score is above the overall average across all cities.

Q7 — Medium | CASE WHEN / Aggregation

Bucket sessions into ETA error bands (e.g., <2 min, 2-5 min, 5-10 min, >10 min, using ABS(actual_duration_minutes - recommended_eta_minutes)) and calculate the acceptance rate within each band.

Q8 — Medium | Window Functions

For each user, use a window function to rank their sessions by session_date and identify the session_id of their very first navigation session (their activation session).

Q9 — Medium | Window Functions

For each session, compute the difference between actual_duration_minutes and recommended_eta_minutes, and use a window function to rank sessions within each city by that error, largest to smallest.

Q10 — Medium | CTE

Using a CTE, first compute each user's total session count and completed-session count, then select the top 20 users by completion rate (minimum 5 sessions).

Q11 — Medium | Funnel Analysis

Using user_events, calculate the stage-to-stage conversion rate through the full funnel: app_open -> destination_search -> route_recommendation_view -> route_selected -> navigation_started -> navigation_completed.

Q12 — Medium | Funnel Analysis / JOIN

For sessions where navigation was started but not completed, identify the most common destination_category and traffic_level combination.

Q13 — Medium | Percentiles

Calculate the 25th, 50th, 75th, and 95th percentile of reroute_count for sessions where navigation_started = True.

Q14 — Medium | Ranking

Rank cities by average ETA error (ABS(actual_duration_minutes - recommended_eta_minutes)) from worst to best, and return only the bottom 5 (worst) cities.

Q15 — Medium | Time-Series

Calculate the daily count of navigation sessions and the daily acceptance rate over the full date range, ordered chronologically.

Q16 — Medium | JOIN / Subquery

For each session, compare the selected route's eta_minutes in route_options to the fastest available option's eta_minutes for that session, and calculate the average 'time cost' of the user's choice versus the fastest option.

Q17 — Hard | Cohort Analysis

Build a signup-week cohort table showing, for each cohort, the retention rate (based on retention_daily.active_flag) at day 7, day 14, and day 30 after signup.

Q18 — Hard | Retention

Using retention_daily, calculate rolling 7-day active users (WAU) and 30-day active users (MAU) for each day in the observation window.

Q19 — Hard | Window Functions / CTE

For each user, compute the number of consecutive days of activity (a 'streak') using retention_daily, and report the longest streak per user.

Q20 — Hard | Statistical / Correlation Proxy

Using SQL aggregate functions, approximate the correlation direction between traffic_uncertainty_score and route rejection by comparing average rejection rate across traffic_uncertainty_score quartiles (use NTILE).

Q21 — Hard | Experiment Analysis

From experiments.csv, calculate route_acceptance rate, navigation_completed rate, and retained_7d rate by experiment_group, along with the absolute and relative lift of Treatment over Control for each metric.

Q22 — Hard | Window Functions

For each user, calculate a running total of navigation_session_count from retention_daily ordered by activity_date, and identify the date each user crossed their 10th completed navigation (successful_navigation_count).

Q23 — Hard | Multi-table JOIN

Combine navigation_sessions and route_options to calculate, for each session, how many toll-free route options were available versus how many were shown with a toll, then relate this to whether the recommended route was accepted.

Q24 — Hard | Segmentation / CASE WHEN

Create a query that segments users into 'High Reroute', 'Occasional Reroute', and 'No Reroute' based on their average reroute_count across all sessions, then compare each segment's 30-day retention using retention_daily.

Q25 — Hard | Advanced Funnel + Cohort

Combine user_events and users to build a cohort funnel: for users who signed up in the same calendar month, calculate what percentage reached each funnel stage (search, route view, route selected, navigation started, navigation completed) at least once within their first 14 days.